Executed n=83 genus-3 image transfer and destination solvers

Exact certificates, four-size cold comparison, and the remaining advantage gate

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Executive result

The pinned degree-three cover from the ordinary binary elliptic curve into a genus-three hyperelliptic Jacobian is now an executed, scalar-preserving transfer. Exact n=83 pullbacks, image order, typed pull/push composition, exceptional fibers, and scalar recovery pass. Named BSGS and rho adapters solve on the actual Mumford image and independently replay every recovered scalar on the source.

The four faithful successful toy sizes are n=9,13,19,23. The preregistered n=7 source-reference exhaustion is retained. Across all measured trials, 400 of 400 destination attempts verified; 320 source attempts at the successful sizes verified, and 80 source attempts at n=7 exhausted. All 720 verified measured records replayed identically.

This establishes a valid transfer and destination solver reachability. It does not establish a weaker destination DLP or a net speedup: both destination methods are generic square-root algorithms, native Cantor work has no certified source-GAE conversion, and wall observations earned only L0 isolation.

Exact transfer certificate

The exact source is `E_0: y^2+x*y=x^3+1` over the polynomial basis defined by `z^83+z^45+z^2+z+1`, with prime subgroup order `r=241785163923079621668568 9`, cofactor four, pinned generator `P`, and public known-answer `Q=[37]P`. The cover is

```
C_0: v^2+u^2*v=u^7+u,
phi(u,v)=(u^3,u*v+1), degree(phi)=3.
```

The final post-panel certificate proves:

- `phi^*(P)` and `phi^*(Q)` are valid nonzero Mumford classes;
- both are killed by prime `r`, so `phi^*(P)` has exact order `r`;
- `[37]phi^*(P)=phi^*(Q)`;
- `phi_*phi^*=[3]` for `P`, `Q`, infinity, and the exceptional target `(0,1)` above the ramified point `(u,v)=(0,0)`;
- `gcd(3,r)=1` and `3^-1 mod r=161190109282053081112379 3`; and
- bound-64 BSGS, without receiving scalar 37, recovers 37 and passes both destination verification and source replay.

The certificate scope is adapter reachability, not a full 81-bit discrete log.

Solver and accounting design

`jacobian_image.rho3` is the generic destination control. `jacobian_image.bsgs` is the generic memory-using candidate. Each cold process constructs the cover, pulls back generator and target,

solves in the image, verifies there, and returns a scalar for independent source verification.

The measurement preserves unlike units. Destination additions/doublings, scalar multiplications, table work, and walk steps are native counters marked uncharged in source GAE. A zero GAE total therefore means “no conversion,” not “free work.” Saved comparisons correctly decline to produce an operation-count speedup.

Measured panel

n	r	source wall ms	destination rho3 native ops / ms	destination BSGS native ops / ms	BSGS/rho native ops	BSGS/source wall
7	29	exhausted	35.475 / 2.288	16.250 / 2.179	0.458	–
9	127	1.857	71.675 / 2.681	32.375 / 2.238	0.452	1.205
13	2,003	1.915	215.150 / 4.369	90.375 / 3.122	0.420	1.631
19	130,873	1.931	1,535.725 / 19.187	592.000 / 9.863	0.385	5.107
23	2,095,853	2.092	6,194.475 / 77.399	2,113.250 / 28.421	0.341	13.586

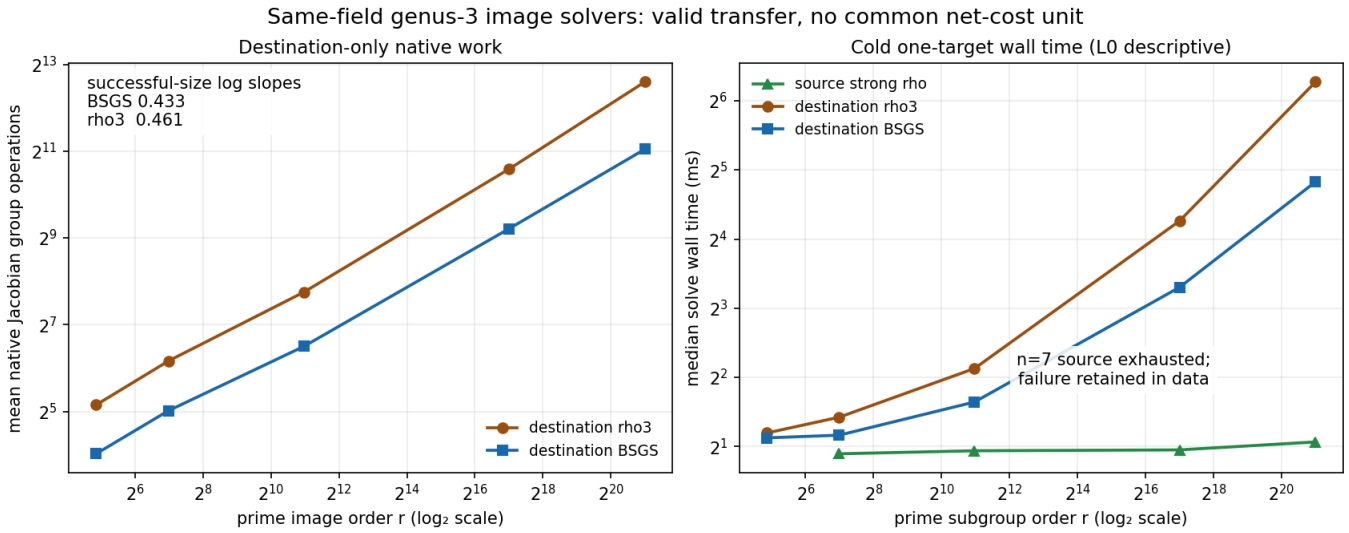


Figure 1: Native destination work and descriptive wall time

BSGS beats rho within the destination by the expected generic time/memory trade. Its successful-size native log slope is 0.433, versus 0.461 for rho3. Neither is evidence of a special Jacobian-image weakness. Descriptive wall time is worse than the source reference and the ratio grows across the panel.

Integrity and chronology

The first strict session start refused before executing workloads because PSI was unavailable. The unchanged frozen spec then ran with `--allow-busy`; all rows earned L0 because isolation could

not be certified. Session ECBS1h2f7b31083d9a wrote all 768 planned records (672 verified, 96 exhausted) and replayed 560 measured verified rows. After a committed follow-up protocol, session ECBS1hddc5cdbe0292 wrote 192 of 192 verified records and replayed all 160 measured rows. Both audits report no problems.

An exact implementation preflight ran before the panels and is retained. The final exact n=83 gate ran after the combined four-size successful panel; the two deterministic certificates are byte-identical. This report treats only the post-panel receipt as the final requested gate.

Verdict and next gate

The v3 verdict is:

Gate	Verdict
valid transfer	supported
destination solver applicability	supported
modeled advantage	unknown
measured advantage	unknown
novel reduction	not applicable

The next discriminating implementation is a complete small-field genus-three index-calculus adapter, not full-order generic n=83 BSGS. It must cover factor base, relation collection, linear algebra, individual logarithm, image access, and source replay. It should first face the existing destination BSGS control at n=9,13,19,23, with native field/memory/matrix counters. Only a lower destination exponent or complete finite-size win should proceed to a common-unit three-arm net comparison under L2 or stronger isolation on an independent host.

Artifact map

- Exact final certificate
- Sealed sessions
- Full run chronology and commands
- Results and measurement guidance
- Machine-readable assessment
- Open exit criteria